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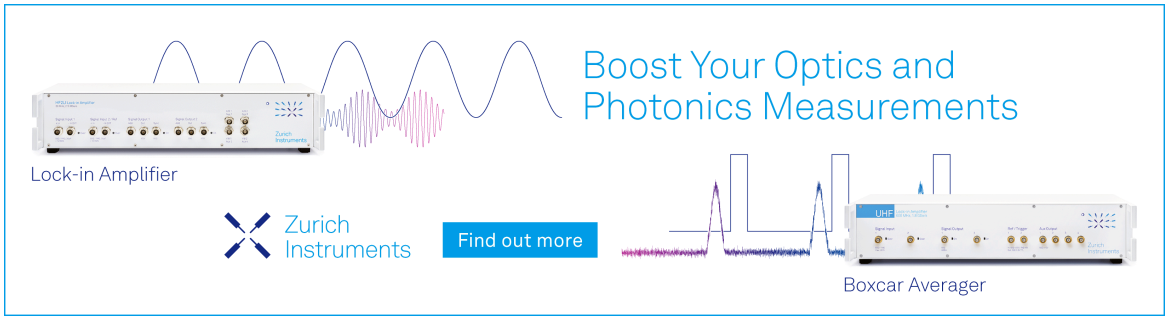

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ABSTRACT

Minimizing the micromotion of a trapped ion in a linear Paul trap is of great importance in maintaining long coherence time as well as implementing quantum logic gates with high fidelity, which is crucial for large-scale quantum computation with trapped ions. Here, by applying the RF (radio frequency)-photon correlation technique, we demonstrate that a machine learning method based on artificial neural networks can quickly search for optimal voltage settings of the electrodes to minimize the trapped ion's micromotion. This machine learning assisted RF-photon correlation technique can be straightforwardly applied to more complicated surface ion traps with many electrodes, where the manual minimization of the excess micromotion generated by stray electric fields would become extremely challenging for the larger number of electrodes with various voltage settings. Instead, the presented machine learning assisted method provides an effective and automatic way to address this need.

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Recently, artificial intelligence (AI) has made great changes to industry, marketing, and social life, such as facile/voice recognition and prediction of consumer behaviors. Due to its great success, machine learning algorithms have been rapidly applied to diverse research subjects,[1] including physics,[2–8] chemistry,[9] biology,[10] astronomy,[11,12] economics,[13] and finance;[14] weather forecast;[15] medical science;[16] material science;[17] and drug discovery.[18] In physics, it is successful in solving problems from atomic, molecular, and optical physics[7,19] to condensed matter physics,[2,4] from high-energy physics[5] to astrophysics.[11,12] It plays a significant role in reducing the complexity of the problem and in optimization of experimental parameters with various mathematical algorithms. Especially, it can reduce a multi-dimensional problem to only a few dimensions, which greatly simplifies the experimental realization of any NP-hard (non-deterministic polynomial-time hard) problem in physics, such as to improve the fidelity of a quantum gate operation using trapped ions.[19]

AI has been applied in trapped ion quantum systems for training of quantum circuits,[20] optimization of qubit readout fidelity,[19,21] optimization of laser control parameters for the simulation of interacting spin models,[22] and so on. Here, we realize an application using artificial intelligence, i.e., minimization of the excess micromotion of a single trapped ion. The excess micromotion causes the ion to oscillate at the radio frequency (RF), resulting in Doppler shifts as the ion moves toward or away from the laser beam. This excess micromotion will reduce the efficiency of both cooling and state detections and could heavily reduce the ion's coherence time. It also deteriorates the harmonic character of the ion motion, making it impossible to construct an ion quantum gate with high fidelity. Therefore, it is important to reduce the excess micromotion to the highest degree possible.

Multiple techniques have been employed in detecting and minimizing the excess micromotion of trapped ions, such as sideband,[23,24] parametric excitation,[25,26] RF-photon correlation,[27,28] etc. Among these, the RF-photon correlation technique[27,28] is one of the most used methods for measuring the amplitude of excess micromotion. It measures the histogram of the emitted ion fluorescence as a function of micromotion index β, laser detuning $\Delta = \omega_L - \omega_0$, and recording time,[28] leading to precise knowledge of the amplitude of excess micromotion. Despite its success, it requires adjusting DC voltages repeatedly. However, direct mathematical relation between DC voltages and the excess micromotion index β is obscure, and the minimization of the micromotion is clearly a multi-dimensional problem, which is a natural playground where machine learning can play a major role.

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In this Letter, we report the application of neural network algorithms to tackle such a multi-dimensional issue, and its extremely effectiveness in minimization of the excess micromotion for the single trapped ion system. One of the most successful machine learning algorithms to attack such a high-dimensional problem is artificial neural network.[29] It performs exceedingly well in a variety of scenarios ranging from image and speech recognition[30] to game playing.[31] Recently, applications of neural networks to the study of physical phenomena have been introduced.[32–35] Although it has been used in only a few cases, it is of fundamental and practical interest to solve such a high-dimensional problem using a neural network-based machine learning method.

We introduce an artificial neural network in search for a relation between a set of voltage parameters U_i and micromotion index β_j. The parameters of the neural network are optimized to fit the experimental data, in other words, trained. We validate the accuracy of this approach by measuring the magnitude of the residual excess micromotion, with the voltage set output from the anneal algorithm applied onto the corresponding electrodes. The power of the neural-network-based machine learning is demonstrated, obtaining the state-of-the-art level of minimization in excess micromotion.[27,28]

The ion trap used in this experiment consists of four gold-plated ceramic blade electrodes, whose edges are parallel to the longitudinal (x) axis of the trap, as shown in Fig. 1(a). Two opposite blades are driven with an RF potential with respect to the other two static blades, creating the transverse (y–z) quadrupole confinement potential. Appropriate static potentials provided by the longitudinal five-segment static blades serve to confine the ion along the x-axis. The RF electric quadrupole potential near the center of the trap is set by the RF amplitude on the trap electrode V_0. As shown in Fig. 1(b), the gap between the electrodes is 470 and 220 μm in the y and z directions, respectively. The distance from the trap center to the electrodes is $R \approx 259\,\mu$m. Other parameters in our blade trap are shown in Table I with a 0.45 W RF power. We use a 10 nF capacitor between two helical coils, which acts as input of the RF oscillating field, thus eliminating the phase difference between two RF electrodes.

We measure the photon arrival time as a histogram using the RF-photon correlation method, from which excess micromotion index β can be obtained by fitting the histogram profile to the following equation:[28]

$$\begin{aligned} S(\Delta, t) &= \left| \int_{-\infty}^{\infty} A(\omega - \omega_0) E(\omega) e^{i\omega t} d\omega \right|^2 \\ &\propto J_0^2(\beta)|A(\Delta)|^2 + J_1^2(\beta)\left(|A(\Delta + \Omega_{rf})|^2 + |A(\Delta - \Omega_{rf})|^2\right) \\ &\quad + 2J_0(\beta)J_1(\beta)|A^*(\Delta)A(\Delta + \Omega_{rf}) \\ &\quad - A(\Delta)A^*(\Delta - \Omega_{rf})|\cos(\Omega_{rf} t + \varphi) \\ &\quad + 2J_1^2(\beta)|A(\Delta + \Omega_{rf})A^*(\Delta - \Omega_{rf})|\cos(2\Omega_{rf} t + \varphi'), \end{aligned} \quad (1)$$

where $A(\Delta)$ is the spectral function of frequency response and $\Delta = \omega_L - \omega_0$ is the detuning of the excitation laser frequency from the atomic resonance. As shown in Fig. 1(c), we excite the trapped ion from $|^2S_{1/2}, F = 0, 1\rangle$ to $|^2P_{1/2}, F = 0, 1\rangle$ with the detection laser $\lambda_1 \approx 369.5$ nm and record the histogram of ion fluorescence using a start–stop time-to-digital converter (HRM-TDC, SensL) to which photon numbers are fed from a photomultiplier tube (H10682–210, Hamamatsu). The repumping laser $\lambda_2 \approx 935.2$ nm is kept on all the

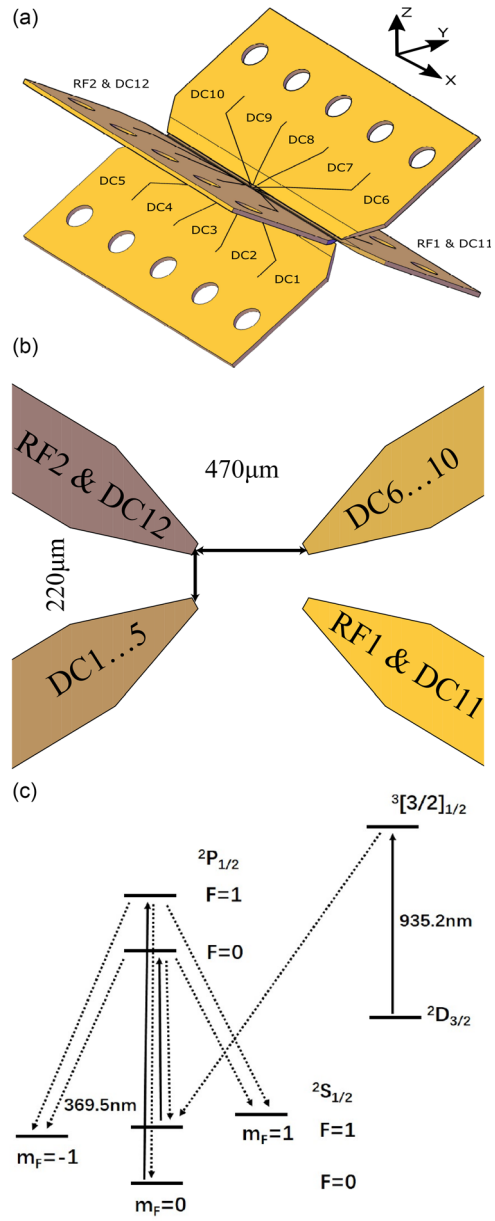

FIG. 1. (a) The five-segment blade ion trap and (b) its cross section in radial direction. (c) The relevant energy levels of $^{171}Yb^+$ ion in the experiment.

TABLE I. Parameters in our blade trap. Unit of electrode voltage is in volt. The DC offset voltages on RF electrode RF1 and RF2 are −0.079 and −0.22 V, respectively. Ω and $\omega_i (i = x, y, z)$ represent the RF frequency and trapping frequency along i-axis, respectively. Unit of frequency is in MHz.

V_{RF1}	V_{RF2}	$V_{DC1,2}$	V_{DC3}	$V_{DC4,5}$	$V_{DC6,7}$
227.84	227.84	11.75	1.742	9.35	12
V_{DC8}	$V_{DC9,10}$	Ω_{RF}	ω_x	ω_y	ω_z
1.9	9.6	22.625	0.744	1.382	1.655

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time during the measurement. Detuning of the detection laser is kept at $\Delta = 2\pi \times -20\,\text{MHz} \approx -1\Gamma$ to obtain a relatively large photon-correlation signal Δ/S_0.[28] The power of the detection laser is kept in the low intensity limit, i.e., the ion is driven below the saturation limit $I/I_{sat} \approx 0.5$. During the experiment, we alternately switch the direction of the detection laser between the x-direction and y-direction in order to measure β_x and β_y, respectively. For each histogram measurement, the data acquisition time is 30 s. The average photon count rate at a detuning of -1Γ is about $27\,000\,\text{s}^{-1}$. The background count rate is $1009\,\text{s}^{-1}$ ($1782\,\text{s}^{-1}$) with x-direction (y-direction) detection laser present.

In our experiment, a progressive algorithm is designed for searching the null point of our blade trap. A pre-written python program carries out the first module for each running cycle in Fig. 2(b) and can update the voltages automatically and in real time to the specified DC electrodes by communication between a control PC and a programmable precision DC voltage power supply (BS1–16-14, Stahl-electronics). Prior to applying the algorithm, we find a set of voltages $V_{i,c}$ through position compensation, where we first ramp up the RF power and keep the ion's position $P(x_0, y_0, z_0)$ with an EMCCD (electron-multiplying charge-coupled devices) camera (iXon 897, Andor) by adjusting voltages through the DC voltage power supply when we ramp down the RF power.

Now, we start to apply the neural network, whose configuration is shown in Fig. 2(a). The experimental input is the applied voltages to the electrodes, and the output is the measured micromotion index β. In between of them are layers of hidden neurons Y_j^i, which can be expressed by

$$\begin{aligned} Y_j^1 &= f_j^1\left(\sum_{n=1} w_{n,j}^1 V_n\right) + b_j^1, \\ Y_j^i &= f_j^i\left(\sum_{n=1} w_{n,j}^i Y_n^{i-1}\right) + b_j^i, \end{aligned} \tag{2}$$

where $f(x)$ is the activation function. Here, we use the Leaky rectified linear unit (Leaky ReLU) function.[36] The output can be expressed by

$$\beta_j = f_j\left(\sum_{n=1} w_{n,j}^{i+1} Y_n^i\right) + b_j.. \tag{3}$$

In Fig. 2(b), the experimental training cycles are illustrated. Each cycle has five modules. In each cycle, a Gaussian distribution of voltages for each compensation electrode is generated first and written into a DC file, by using $V_{i,c}$ as the center and ΔV_i as the standard deviation. The range of this distribution is limited by the stable region based on the Mathieu equation. Then, these voltage sets are applied to the electrodes one at a time through programmable precision DC voltage power supply, and their corresponding histograms of ion fluorescence in the x and y directions are measured, respectively. Each histogram is taken three (or five) times to minimize the effects of drifting laser frequency and fluctuating laser power as well as the RF noise. Subsequently, the micromotion indexes β_x and β_y for each voltage set are obtained by fitting each histogram to the Eq. (1).

Using these voltage sets and their indices, an initial model of the neural network can be constructed. Usually, the stochastic gradient descent (SGD) algorithm[37,38] is used for such a purpose. In order to build a good model, the loss function has to be as small as possible,

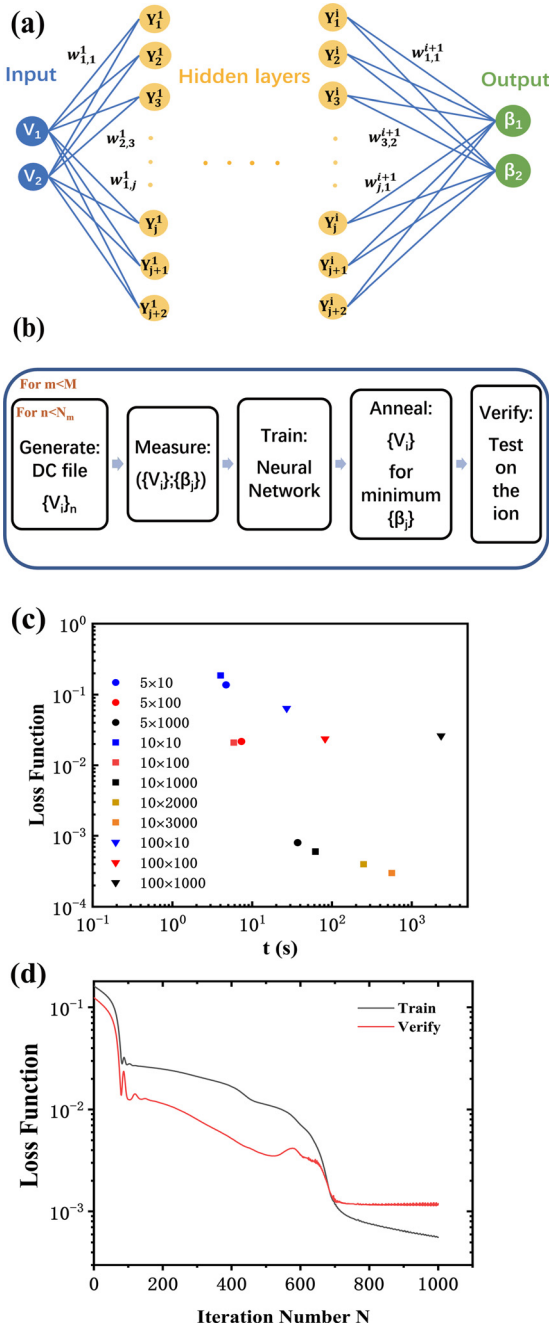

FIG. 2. (a) The neural network in our configuration. (b) The schematics of training cycles for the minimization of micromotion index β_x, β_y using neural-network-based machine learning. Each cycle has five modules. (c) The loss function that can be achieved by the training measurement and the corresponding training time for given layers × number of neurons in each layer. Noted here that the number of iterations is 1000 for all cases. The circles, squares, and down-triangles represent 5, 10, and 100 layers, respectively. Blue, red, black, brown, and orange correspond to 10, 100, 1000, 2000, and 3000 neurons per layer, respectively. (d) The variation of the loss function of β_y with iteration number N. Black curve represents the loss function given by the training measurement, while the red curve represents the loss function given by the verification measurement.

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$$L(\beta) = MSE(\tilde{\beta}, \beta) = \frac{1}{n}\sum(\tilde{\beta}_i - \beta_i)^2. \quad (4)$$

In the third module of Fig. 2(b), where the neural network algorithm is applied, we employ the backpropagation (BP) algorithm[39] to realize Adam (an variation of SGD) in order to iterate the neural network until $L(\beta)$ becomes small enough in the neural network.[40] Consequently, appropriate expression for each neuron with other neurons can be precisely determined by giving the corresponding bias and weight, such that output neurons can be calculated for any input set. Thus, we have established a well-working model of $(V_i; \beta_i)$. The following annealing process gives us the optimal voltage set $V_{i,m=1}$ allowing for the minimum β_x and β_y straightforward, where the subscript m represents the optimization after m experimental cycles. In order to verify the results, an additional test is performed with multiple voltage sets in the vicinity of $V_{i,m=1}$.

Since the range of the voltage sets in first cycle is usually quite large, in order to have a precise target voltage set and a better minimization of the excess micromotion, we repeat the process in the second cycle. A new Gaussian distribution of voltages is generated for each compensation electrode using $V_{i,m=1}$ as the center and $\frac{\Delta V_i}{2}$ as the standard deviation, and the previous process is repeated again. In the experiment, we continue such cycling until the voltage set converge to the same result $V_{i,n}$. By doing this, the optimal voltage set corresponding to the actual null point is reached after n cycles, leading to minimum values of β_x and β_y.

In the first experiment, we test our model with two-input and two-output, where two inputs are voltages V_2 and V_7 on DC2 and DC7, respectively, and two outputs are β_x and β_y, respectively. We choose number of layers and neurons per layer by comparing the value of the resulted loss function and the corresponding training time of model, as shown in Fig. 2(c). In our trial with fixed number of iterations $N = 1000$, the loss function typically reduces with the increase in neurons per layer for the same number of layers (five or ten), indicating the accuracy of the neural network increases with increasing neurons per layer, while the corresponding training time also increases. However, for the number of layers equals 100, the loss function first reduces to approximately 0.02 when neurons per layer increases to 100, and then it drops to a local minimum and stops decreasing when neurons per layer are further increased. In our experiments, the loss function is chosen to be on the order of 10^{-4} at which the model is fully trained. Obviously, 100 layers does not meet our criteria. Therefore, we choose ten hidden layers, each having 1000 neurons

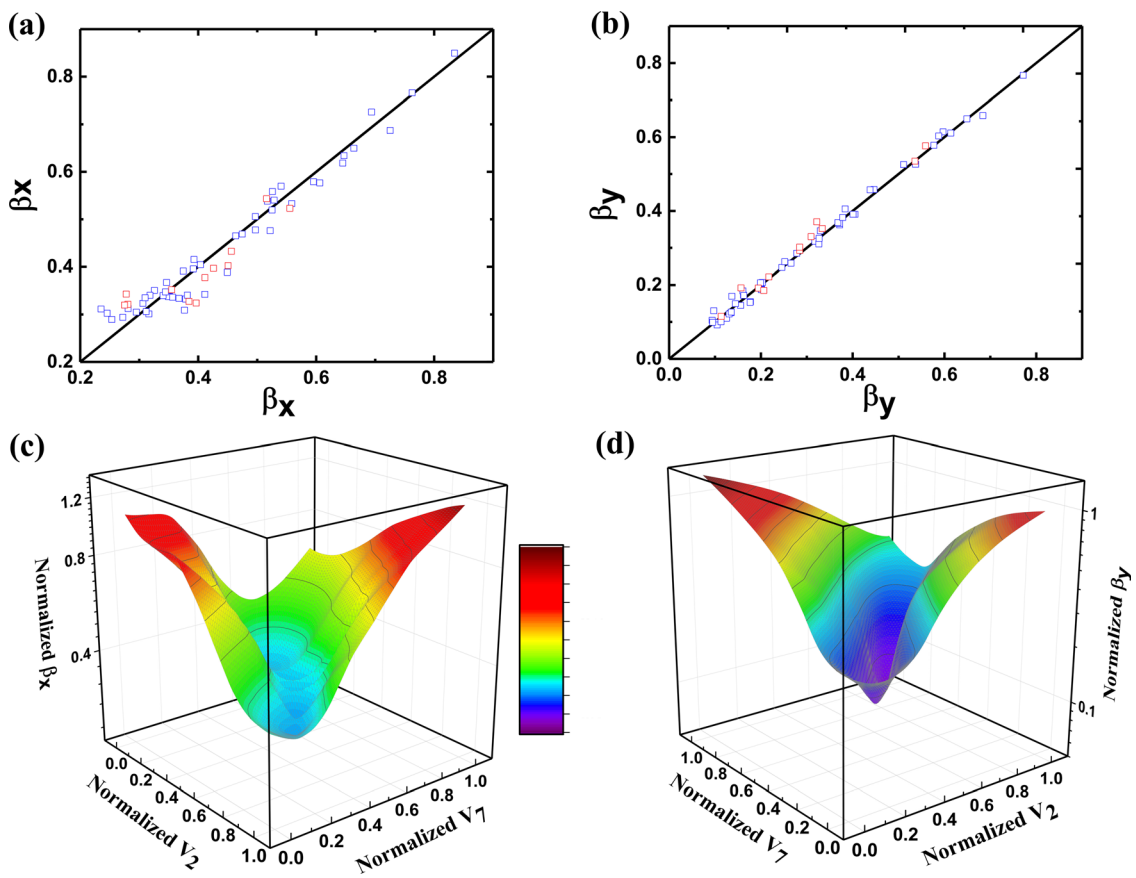

FIG. 3. The distribution of (a) β_x and (b) β_y from prediction (blue squares) and verification (red squares) as a function of ground truth (black line) using the experimental measurements as the training set. Black line represents experimental data, while blue and red squares correspond to predication by the model and further test of the model with the measurement in the last module of each block in experimental procedure. The three-dimensional view of the learning model for (c) normalized β_x and (d) normalized β_y as the function of V_2 and V_7, respectively.

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taken both resulted the loss function and training time into account. Further increase in neurons per layer than what we used can lead to a few possible outcomes: it would inevitably need larger computational resources and longer time in calculation and iteration; in addition, it can easily lead the loss function to a local minimum much larger than the desired value.

The output sequence of $\beta_x(V_1, V_2, ..., V_i)$ and $\beta_y(V_1, V_2, ..., V_i)$ from our model is then compared with the measured sequence $\tilde{\beta}_x(V_1, V_2, ..., V_i)$ and $\tilde{\beta}_y(V_1, V_2, ..., V_i)$ over the whole parameter space to evaluate $L(\beta)$. Once it reaches a value that is smaller than a threshold ε, we conclude that the model is appropriate. So, it can be taken as a parameter for verifying the robustness of the model. Figure 2(d) demonstrates the variation of the loss function with iteration number N during the training for β_y. Black curve represents the loss function between measurement and the prediction from the model, while the red curve represents the loss function between verification measurement and the prediction from the model. As is shown, the loss function drops logarithmically with the iteration number. As is evident in Fig. 2(d), both of them rapidly decrease to 0.001 after roughly 700 iterations, indicating that the constructed model has been well-trained for $N \geq 700$.

This is double-confirmed in Figs. 3(a) and 3(b).[41] The measured β used for training, the predicted β by the trained model, and the measured β based on trained model for its verification are all plotted. For both β_x and β_y, the predicted value (blue square) and the verify data (red circle) are in close proximity with the measurement (black line), proving the accuracy, effectiveness, and robustness of the model. The output from the trained model can be visualized by Figs. 3(c) and 3(d), which show the three-dimensional view for β_x and β_y as the function of normalized V_2 and V_7, respectively. It is worthy to notice; in the investigated parameter space of V_2 and V_7, both β_x and β_y have a valley shape, indicating the existence and the location of a combined minimum β_x and β_y.

We further extend such tests to a setting with three input voltages (V_2, V_3, and V_7) and four input voltages (V_2, V_3, V_7, and V_8). For three input voltages, a similar training to the model using neural network has been performed. As shown in Figs. 4(a) and 4(b), the predicted value (blue square) and the verify data (red circle) agree well with the measurement (black line), demonstrating its accuracy, effectiveness, and robustness.

For the four input voltages, we also find the global minimum of β_x and β_y for a specific combination of (V_2, V_3, V_7, and V_8). The resulted histograms of the ion fluorescence corresponding to minimum excess micromotion is illustrated in Figs. 5(a) and 5(b). They show histograms of the fluorescence in two periods along the x- and y-directions, respectively, where each bin corresponds to 216 ps. The corresponding β_x and β_y are inferred to be $\beta_x = 0.0060 \pm 0.0026$ and $\beta_y = 0.0308 \pm 0.0028$, respectively. Such orders of magnitude prove an excellent minimization and are comparable to Refs. 27 and 28, clearly demonstrating the efficacy of machine learning in this application.

In summary, we have demonstrated a high-level of micromotion suppression by neural-network-based machine learning, using the RF-photon correlation technique. With our approach, β_x and β_y have been reduced to 10^{-3} and 10^{-2} orders of magnitude, respectively, reaching the state-of-the-art sensitivity and comparable to those used for atomic clocks[27,28] and quantum metrology. β_z is estimated to

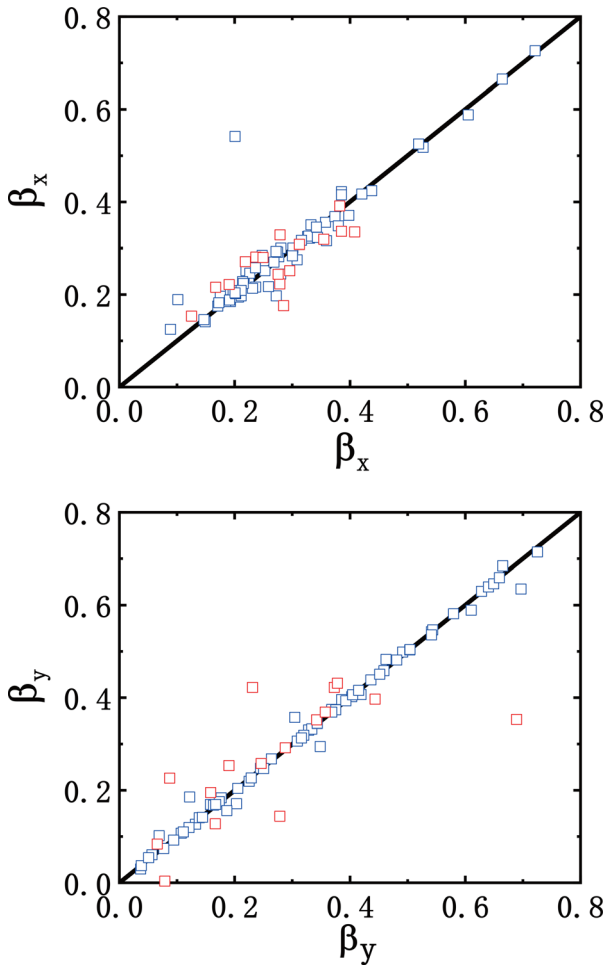

FIG. 4. The distribution of (a) β_x and (b) β_y from prediction (blue squares) and verification (red squares) as a function of ground truth (black line) for three input voltages (V_2, V_3, and V_7). The black line represents experimental data, while blue and red square correspond to predication by the model and further test of the model with the measurement in the last module of each block in experimental procedure.

within 10^{-2} according to its relation with β_y. This method is highly automatic and can be easily applied to surface ion traps and proposed ion-trap quantum computer with large scale of integrated ion traps where there are tens or even hundreds of different DC electrodes while manual minimization of excess micromotion generated by stray electric fields would become extremely challenging for both modeling and implementation. Moreover, the reported algorithm is applicable to a broad range of real applications of trapped-ion experiments, such as machine learning assisted electric field sensor.

Furthermore, the demonstrated ability would enable us to further investigate one of the most fundamental research in quantum mechanics, the motion of a driven harmonic oscillator in mixed external fields, such as sensitive measurement of extremely small displacements and ultraweak forces.[42–45] We also envision that this approach can be generally applied to other similar scenarios in atomic,

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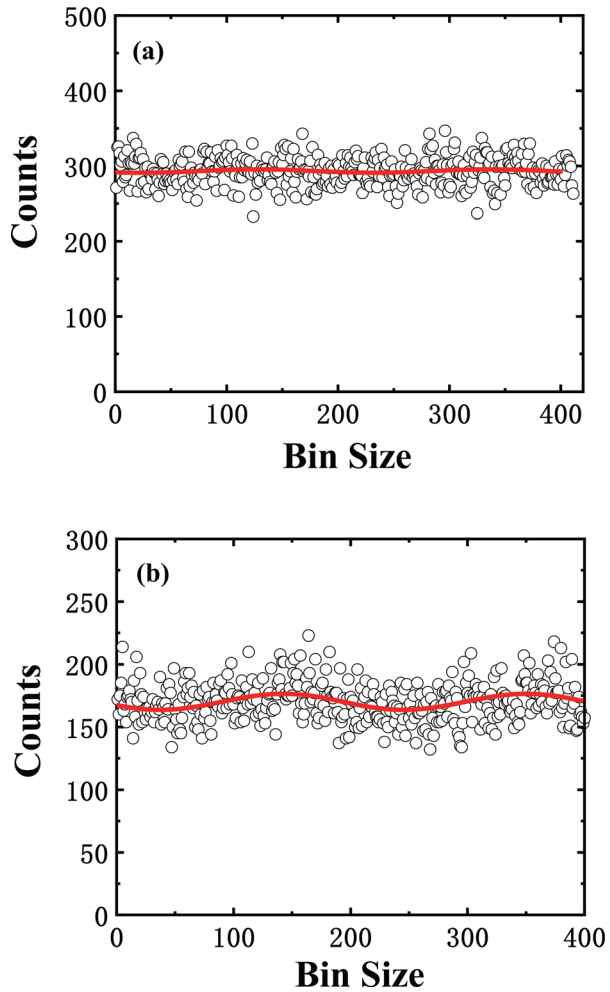

FIG. 5. The obtained best minimization of micromotion with (a) $\beta_{x,min} = 0.0060 \pm 0.0026$ and (b) $\beta_{y,min} = 0.0308 \pm 0.0028$ during the photon-correlation measurement with our machine-learning approach. Black circles represent raw experimental data, while red line corresponds to the fitted spectrum according to the Eq. (1). Each bin corresponds to 216 ps.

molecular, and optical physics, quantum information processing, and quantum computation.

AUTHORS' CONTRIBUTIONS

Y.L., Q.-f.L., and P.-f.L. contributed equally to this work.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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